BorrowBox — Complete Developer Guide & Reference

Full-Stack Equipment Rental Marketplace | Commands & Database Documentation

PART 1: TERMINAL & DEVELOPER COMMANDS

• Start Backend Development Server

Starts Spring Boot application on http://localhost:8080.

```
cd backend
mvn clean spring-boot:run
```

• Run Unit & Concurrency Tests

Executes JUnit 5 test suite verifying PricingEngine and BookingConcurrency.

```
cd backend
mvn test
```

• Build Production JAR Package

Compiles source code and builds target/borrowbox-backend-1.0.0.jar.

```
cd backend
mvn clean package -DskipTests
```

• Set MySQL Environment (PowerShell)

Configures external MySQL connection credentials.

```
$env:DB_URL="jdbc:mysql://localhost:3306/borrowbox_db?useSSL=false&allowPublicKeyRetrieval;=true&serverTime
zone;=UTC"
$env:DB_USERNAME="root"
$env:DB_PASSWORD="your_password"
```

PART 2: DATABASE SCHEMA & TABLES (13 TABLES)

Table Name	Description & Function	Key Attributes
users	User account credentials & roles	id, full_name, email (UNIQUE), role, password_hash
categories	Equipment categories	id, name, slug (UNIQUE), icon_name
equipment	Gear listings & pricing	id, owner_id (FK), category_id (FK), daily_price, security_deposit, status
equipment_images	Product gallery photos	id, equipment_id (FK), file_path, is_primary
bookings	Rental reservations	id, booking_reference, start_date, end_date, total_amount, booking_status
booking_status_history	Audit log for state changes	id, booking_id (FK), previous_status, new_status, notes
condition_reports	Pre & post rental inspection	id, booking_id (FK), inspection_stage, condition_notes
reviews	Verified rental ratings	id, booking_id (FK), rating (1-5), comment
disputes	Moderation tickets	id, booking_id (FK), reason, description, status

PART 3: DOCUMENTED DEMO ACCOUNTS

Role	Email Address	Phone Number	Password
Administrator	admin@borrowbox.com	+91 98765 00001	Password123!
Equipment Owner 1	rahul.owner@borrowbox.com	+91 98765 11111	Password123!

Equipment Owner 2	priya.owner@borrowbox.com	+91 98765 22222	Password123!
Renter 1	alex.renter@borrowbox.com	+91 98765 33333	Password123!
Renter 2	sneha.renter@borrowbox.com	+91 98765 44444	Password123!